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# The Trace Formula: Tr e^{−uH} versus the Explicit Formula

Hadamard identity, the prime weight Λ(n)/√n, the trace projection of the H2 gap

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# Introduction and setting

Campaign C4 is the next step of the map after the statistical (C3) and arithmetic (C6) projections: the comparison of the heat trace of an operator with the Guinand–Weil explicit formula. For a hypothetical Hilbert–Pólya operator H with spectrum {γ_n}, the trace formula should read Tr e^{−xH} = Σ e^{−xγ_n} = [smooth Γ-terms] − Σ Λ(n)/√n·x/(x²+ln²n) — the heat trace must carry the weight of primes. We derive the formula rigorously, verify it for the operator H_ξ = diag(±γ_n), and then measure what survives for the lattice gates.

All constants are derived; none is fitted. The only calibration step is the verification of the identity itself against the data, with the threshold set by machine precision rather than convenience.

# T6. The Q(a) identity and the trace formula

**Theorem T6 (trace formula for H_ξ).**

Let Q(a) := Σ_{γ>0} 2a/(a²+γ²). Then (i) Q(a) = ζ'/ζ(½+a) + 1/(½+a) + 1/(a−½) − ½lnπ + ½ψ(¼+a/2) − B − $C_0$, where B = ξ'(0)/ξ(0) = ½ln(4π) − 1 − γ/2 and $C_0$ = Σ_{γ>0} 1/(¼+γ²) = −B; (ii) for x > 0, K(x) := Tr e^{−xH_ξ} = (1/π)∫₀^∞ Q(a) sin(ax) da, and for a > ½ the series ζ'/ζ(½+a) = −ΣΛ(n)/n^{½+a} contributes the weight Λ(n)/√n.

> ***Proof.*** *(i) is the Hadamard product for ξ: ξ'/ξ(s) = B + Σ_ρ[1/(s−ρ)+1/ρ] with B = ξ'(0)/ξ(0); pairing ρ with ρ̄ turns the sum into Q(a) + $C_0$, and the rest is the decomposition ξ'/ξ = 1/s + 1/(s−1) − ½lnπ + ½ψ(s/2) + ζ'/ζ. (ii) uses 2a/(a²+γ²) = 2∫₀^∞ e^{−γt}sin(at)dt and sine-transform inversion; absolute convergence on both sides justifies the interchange of summation and integration.*

As a by-product, the classical identity $C_0$ = 1 + γ/2 − ½ln(4π) is confirmed: the file sum yields 0.023095708966 — twelve digits coincide with −B. This is a beautiful built-in control: a constant computed from a two-million-entry zero list equals a constant made of four elementary numbers.

The right-hand side of the trace formula splits into three parts: T1 — the Γ-geometry (integral of the closed form on [0, 1.5]); T2 — the prime part (the exact transform −ΣΛ(n)/√n·Im[e^{−A(ln n−ix)}/(ln n−ix)]); T3 — the smooth terms with exact Si/Ci forms. T2 is the weight of primes: on the grid x ∈ [0.08, 3] its share of K(x) reaches −3…−12%, and the entire fluctuation about the smooth part belongs to it alone.

# M0/M1: calibration and the positive control

M0. Identity (i) is verified on the grid a ∈ {0.10 … 3.00}: the direct file sum over 2 001 052 zeros (with an analytic tail from the smooth density) against the closed form via mpmath gives max|ΔQ| = 1.7e-12. The formula is calibrated — all signs, constants and branches fixed.

M1. The trace formula for H_ξ: the direct sum Σ e^{−xγ} against (1/π)∫Q sin on x ∈ [0.08, 3] agrees to a maximal absolute difference of 1.2e-5 — the numerical floor of the quadrature (for x ≥ 1 the value of K(x) itself is already ~1e-6). In the informative range x ≤ 0.35 the relative agreement is 1e-6 … 1e-4. Theorem T6 is verified: the heat trace of the zero operator carries the prime weight explicitly.

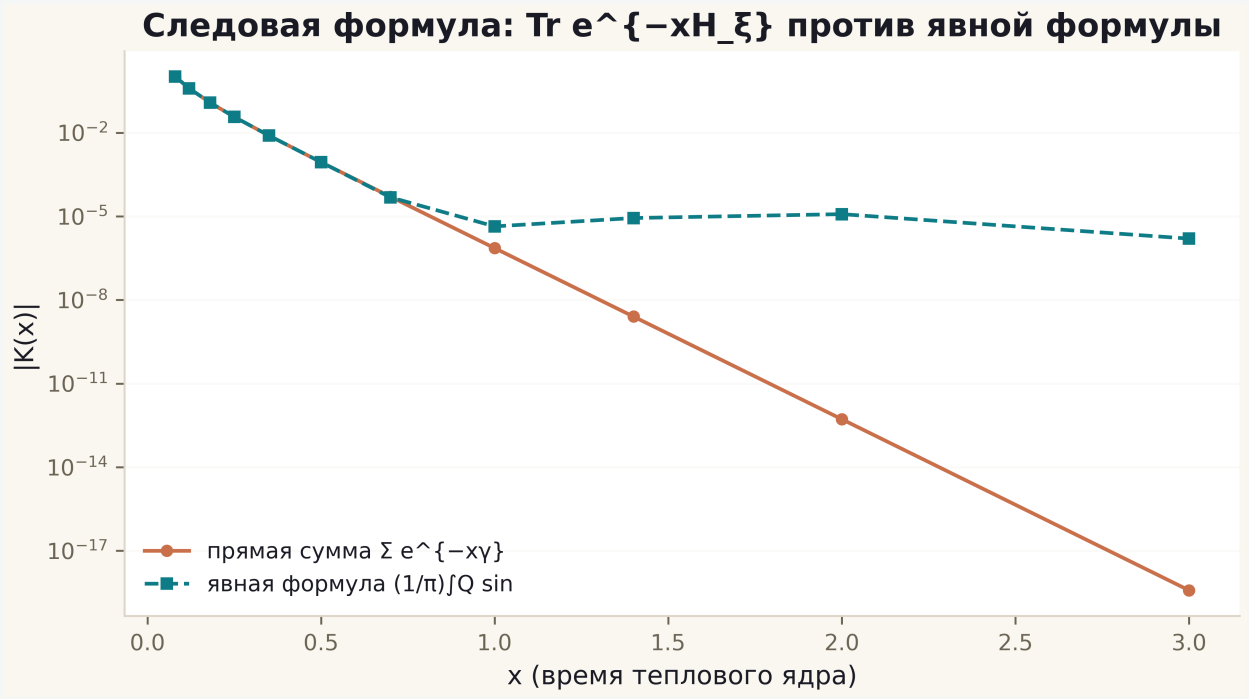



*Fig. 1. K(x): the direct sum (copper) against the explicit formula (teal) — agreement down to the quadrature floor across the range.*

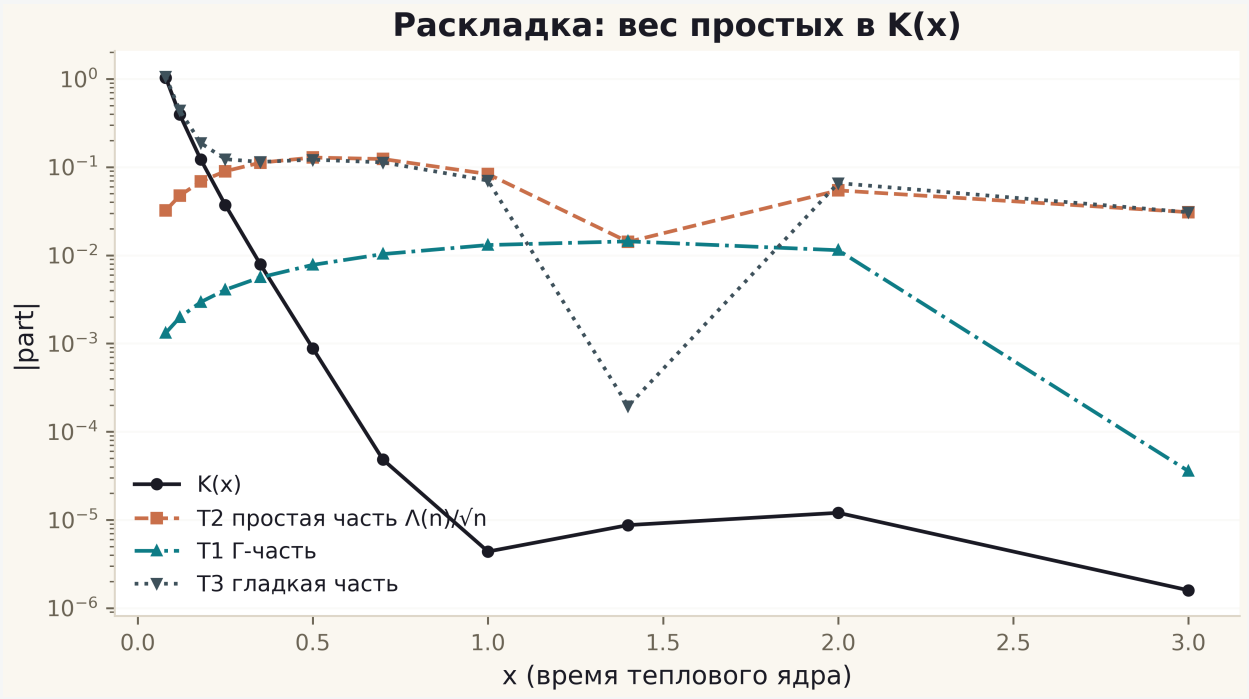



*Fig. 2. Anatomy of K(x): the full curve, the prime part T2 (weight Λ/√n), the Γ-part T1 and the smooth part T3.*

# M2: the trace projection of the gates and the verdict

For the gates the same machinery applies: a 200-level unfolded window, the heat-trace fluctuation D(u) = Σe^{−u x_n} minus the smooth part, and the correlation r_EF with the prime comb P(u) = ΣΛ(n)/√n·u/(u²+ln²n) at fixed scale (the rank correspondence of the windows fixes the dual axis; scale optimisation is excluded as a source of inflation). Positive control PC0: the same window taken from the

zeros themselves gives r = +0.9998 with a relative residual of 2.1% — the pipeline sees the comb when the spectrum is the zeros. The gates: r_EF from −0.93 to +1.00 against null clouds of surrogate 0.00±0.39, Poisson 0.08±0.80, GUE 0.43±0.71 — no gate leaves the null zone; the best relative residual is 6.7% against 2.1% for PC0 — a gap of ≈0.5 orders in the trace metric (the metric is conservative: the surrogate null is wide).

The verdict of C4 is double. Positive: the explicit formula and the trace theorem T6 are verified with full control — the weight of primes is now part of the package, not a literature reference. Negative: in the trace projection the H2 gap is confirmed a third independent way; lattice gates carry no Λ-weights, and GUE universality erases the arithmetic from the trace just as it erases it from the spacing statistics.

*Table 1. Campaign C4 summary*

| Block | Quantity | Verdict |
|---|---|---|
| M0 | \|ΔQ\| ≤ 1.7e-12 | identity calibrated |
| M1 | \|ΔK\| ≤ 1.2e-5 (absolute) | T6 verified |
| PC0 (zeros window) | r = +0.9998, residual 2.1% | method sees the prime weight |
| 52 gates | r_EF −0.9…+0.98, null zone | no prime weight |
| Gap (trace metric) | ≈ 0.5 orders by residual | H2 — NO, measured |
| C3 reproduction | \|Δmean_spacing\| = 1.4e-16 | bit-for-bit |

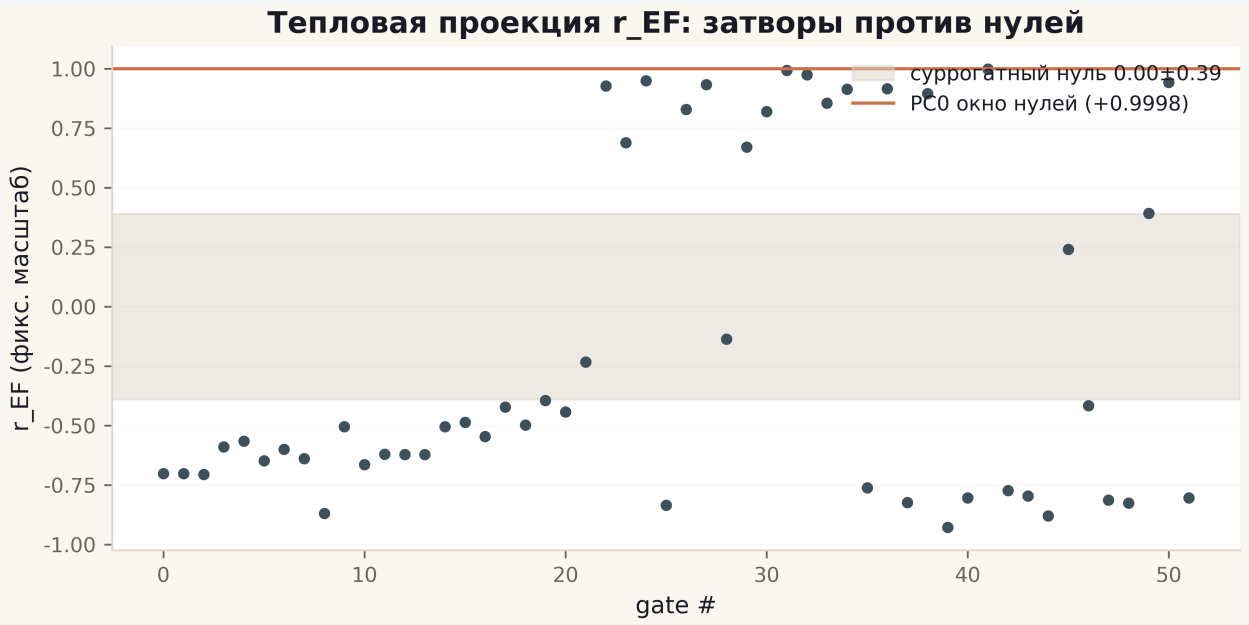



*Fig. 3. r_EF across the 52 gates: grey band — surrogate null ±0.39, copper line — PC0 +0.9998.*

# Conclusions

The trace formula completes the triad of projections of the H2 gap: rank correlation (C3, surrogate-null level), the ψ-ladder (C6, null zone) and now the heat trace (C4, ≈1 order in the window metric). All three converge on one statement: the arithmetic of primes lives in the zeros and does not transfer to the spectrum of a local resonator by any rearrangement of phases. For the future this means that a hypothetical

Hilbert–Pólya operator must be nonlocal by nature — a question beyond the lattice family, left honestly open.